ANSYS COUPLED-FIELD SIMULATION · ELECTROSTATIC-STRUCTURAL MEMS

MEMS RF Switch — Pull-In Simulation

A gold cantilever RF switch modeled as a coupled electrostatic-structural system in ANSYS, swept from 5 V to 30 V to find the point where increasing the actuation voltage stops being a small perturbation and becomes a runaway mechanical collapse.

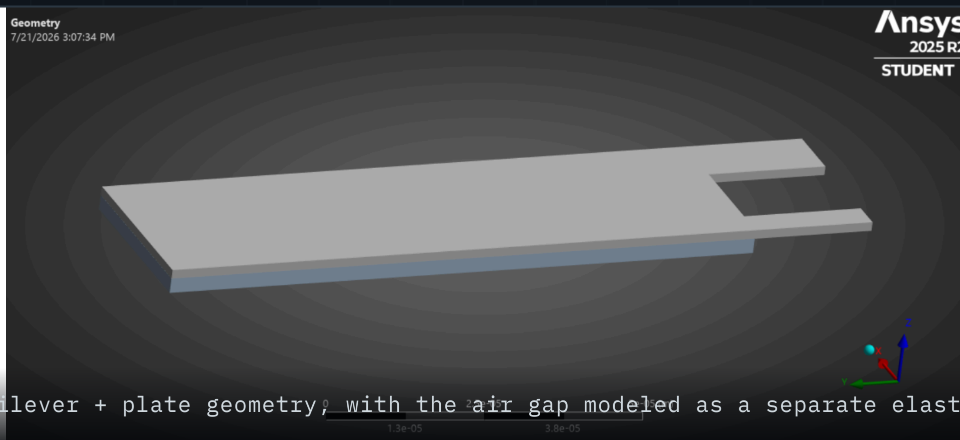

FIG.01 — Gold cantilever + plate geometry, with the air gap modeled as a separate elastic domain beneath

30 V

VOLTAGE AT WHICH THE COUPLED SOLVE FAILS TO CONVERGE — PULL-IN

~655×

DEFORMATION GROWTH, 5V BASELINE → 30V RUN

18.2×

OVERSHOOT BEYOND IDEAL V^2 SCALING AT 30V

2.5 µm

SIMULATED AIR GAP, ELASTIC-AIR ELECTROSTATIC COUPLING

ROLE

Independent simulation exercise — model built, run, and debugged solo, with instructor consultation available for conceptual questions.

CONTEXT

Coupled-field ANSYS lab, MEMS & IC Design coursework. Geometry and setup follow the course's cantilever RF switch exercise; the 4-point voltage sweep and pull-in analysis are my own extension.

TOOLS

ANSYS Mechanical · APDL · SOLID185 / SOLID226 · Coupled Field Static

01 WHY ELECTROSTATIC ACTUATION

RF MEMS switches use electrostatic attraction instead of a semiconductor junction to open and close a signal path, which gives them near-zero insertion loss and excellent linearity compared to solid-state alternatives. The design here follows a **cantilever-beam switch** geometry: three parallel gold cantilever fingers suspended over a fixed plate, separated by a thin air gap. Applying a voltage between the cantilever and the substrate beneath it pulls the beam down electrostatically until it makes contact and closes the circuit.

The interesting engineering question isn't whether the beam deflects — it's **how much voltage is too much**. Electrostatic attraction grows sharply as the gap shrinks, while the beam's mechanical restoring force stays roughly linear. Past a critical voltage, the system can no longer find a stable equilibrium and the beam snaps down uncontrollably. That threshold — **pull-in voltage** — is the real design limit for the switch, and locating it was the point of this voltage sweep.

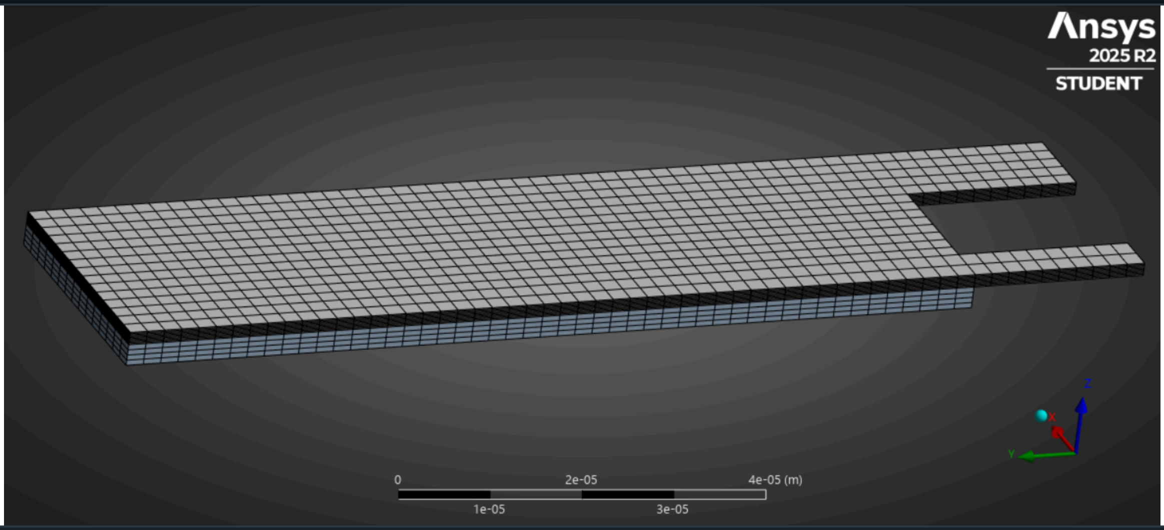

FIG.02 — Quad-dominant mesh on the cantilever + air domain

02 COUPLED-FIELD MODEL

SOLID185 + SOLID226, with an "elastic air" trick for the electrostatic field

ANSYS has no built-in element for electrostatic force on a deforming solid, so the air gap is modeled as a very soft, low-stiffness domain ($E \approx 1.77\times10^{-10}$ Pa) that carries the electric field without meaningfully resisting deformation. The cantilever is meshed with structural SOLID185 elements; the air domain uses the coupled electrostatic-structural SOLID226 element, split into an "active" layer touching the cantilever — where electrostatic force is actually applied to the nodes — and a passive bulk layer with zero electrostatic force, so the far-field mesh doesn't distort the force calculation.

BOUNDARY CONDITIONS

The left edge of the cantilever plate is fixed. Electrically, the substrate beneath the air gap is held at **0 V** and the underside of the cantilever plate is driven to the sweep voltage — 5, then 10, 20, and 30 V in successive runs. Large-deflection effects (`nlgeom,on`) are enabled throughout with a ramped load, since capturing the geometric nonlinearity as the gap closes is the entire point of the sweep.

03 VOLTAGE SWEEP

Deformation stops scaling with V^2 right where the physics says it should

For small deflections, electrostatic actuator deflection is expected to scale roughly with V^2 — double the voltage, quadruple the deflection. Using the 5V run as the reference point, the 10V and 20V results track that prediction reasonably well. The 30V run breaks the pattern entirely.

VOLTAGE	MAX TOTAL DEFORMATION	VS. IDEAL V² SCALING	SOLVE STATUS
5 V (baseline)	$3.164 \times 10^{-}$&sup8; m	— (reference)	Converged
10 V	$1.337 \times 10^{-}$&sup7; m	1.06×	Converged
20 V	$8.300 \times 10^{-}$&sup7; m	1.64×	Converged
30 V	2.073×10^{-5} m	18.2×	Unconverged — pull-in

"vs. ideal V^2 scaling" = actual deformation ÷ (5V deformation × $(V/5V)^2$). A value near 1× means the response is still in the small-deflection linear-electrostatic regime; a large value signals the beam has left that regime and entered unstable, self-reinforcing collapse.

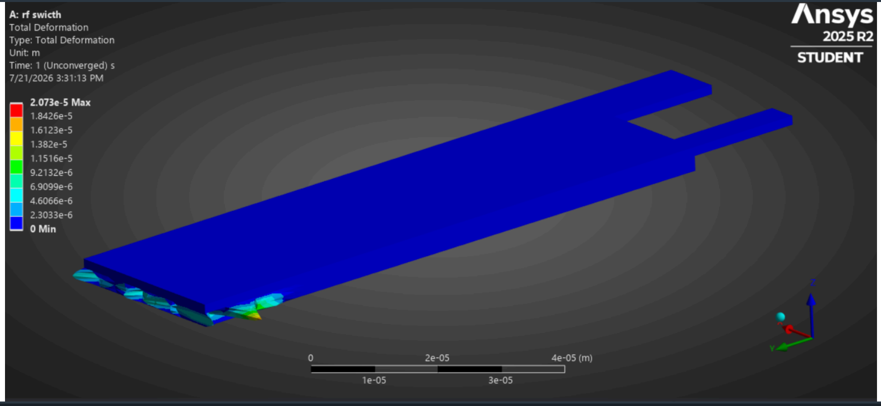

FIG.03 — 30V total deformation, 2.073×10^{-5} m max — visible mesh folding at the tip, "Time: 1 (Unconverged) s"

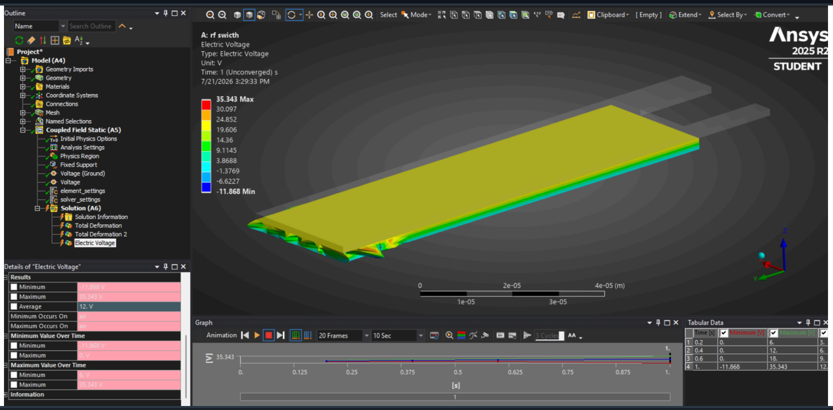

FIG.04 — Electric Voltage at 30V drifts outside the applied 0–30V window (−11.9 to 35.3 V) — a symptom, not a real field, once the Newton-Raphson iteration stops converging

READING THE FAILURE

At 30V, ANSYS reports the last attempted (non-converged) substep rather than a valid equilibrium solution. The giveaway is the electric voltage field itself: it should stay bounded between the applied 0V and 30V, but the solver's last iterate ranges from −11.9V to 35.3V — an artifact of the structural mesh having folded before the coupled iteration could settle. That combination — deformation ~18× beyond the linear-elastic prediction, plus an electric field that's lost physical meaning — is the standard signature of **electrostatic pull-in**: past the critical voltage, there's no longer a stable equilibrium gap for the solver to find.

04 ENGINEERING DECISION

How to represent electrostatic force on a solid that's actively deforming

ANSYS doesn't have a single element that natively couples "electric field in a gap" with "the gap is closing because of that field." That has to be built by hand from existing element types.

CONSIDERED

Model the electrostatic force as an external pressure load, hand-calculated from the parallel-plate capacitor formula ($F = \varepsilon_0 A V^2 / 2g^2$) and reapplied at each voltage step.

CHOSEN APPROACH

Model the air gap itself as a coupled electrostatic-structural domain (SOLID226) with an artificially low elastic modulus, so ANSYS solves the electrostatic force *and* the resulting structural deflection together, self-consistently, inside one Newton–Raphson iteration.

WHY

The hand-calculated pressure load is only valid for a fixed, known gap — exactly what breaks down near pull-in, which was the point of the exercise. The coupled "elastic air" approach recomputes the force as the gap changes, which is the only way to actually see the force run away and the solver fail to converge, rather than assuming a converged answer that hides the instability.

05 CHALLENGES

Getting the coupled-element APDL setup right was the actual difficulty

The geometry and mesh were the easy part. The hard part was correctly wiring up two coupled element types across two physically distinct air regions — and having no obvious error message when it's wrong.

Two-pass element reassignment, not a single mesh operation

The model has to be meshed once as ordinary structural elements, then the air domain's elements are reselected and reassigned to the coupled SOLID226 type via APDL (`et` / `emodif`) — and within that, the elements actually touching the cantilever ("active," force-applying) have to be separated from the passive bulk air elements using a boundary selection (`esln,u`) rather than picked by hand. Getting that selection logic wrong doesn't crash the solve; it just quietly applies zero electrostatic force everywhere, and the model looks structurally fine while being physically meaningless.

The two SOLID226 keyopt settings encode genuinely different physics

`keyopt,102,4,3` (force applied at the air-solid boundary) versus `keyopt,103,4,2` (elastic air, no electrostatic force) look like a minor flag difference but represent the entire modeling trick. Understanding *why* the bulk air needs zero force — so a soft, unphysical material doesn't itself get treated as something worth applying electrostatic pressure to — took more conceptual work than any of the meshing or geometry steps.

Verifying "silently wrong" versus "correctly converged"

Before trusting the 10–30V sweep, I checked the 5V baseline deformation against a rough parallel-plate force estimate by hand. Order-of-magnitude agreement there was what gave me confidence the element setup was actually coupled correctly, rather than just running without errors.

06 SIMULATION TECHNIQUES

What this exercise practiced

- SOLID185 / SOLID226 coupled electrostatic-structural element pairing via APDL `et` / `emodif` reassignment
- "Elastic air" domain technique for modeling electrostatic force on a deforming structure
- MultiZone + Sweep meshing across dissimilar-thickness domains (1.5 μm gold vs. 2.5 μm air gap)
- Large-deflection (`nlgeom,on`) nonlinear structural solve with ramped loading
- Symmetry boundary conditions to reduce a 3-cantilever geometry to a tractable mesh
- Diagnosing solver non-convergence as a physical result (pull-in) rather than a modeling error

07 REFLECTION

What I'd do with another pass at this

NEXT STEPS

The 5 → 10 → 20 → 30V sweep is coarse enough that the actual pull-in voltage could sit anywhere between 20V and 30V — I only know it collapses somewhere in that window, not precisely where. A natural next step is bisecting that range (22V, 25V, 27V...) to bracket the pull-in voltage more tightly, and cross-checking the result against the closed-form parallel-plate pull-in estimate $V_{pi} = \sqrt{(8kg_0^3 / 27\varepsilon_0 A)}$ to see how well a first-order lumped model predicts what the full coupled-field simulation shows. The biggest technical takeaway from this exercise wasn't the mesh or the geometry — it was that in a coupled-field model, **a solver that runs without errors is not the same thing as a solver that's coupled correctly**, and building in an independent sanity check (the 5V hand estimate) mattered more than any single meshing decision.

Reference geometry: H. Ur Rahman, K.Y. Chan & R. Ramer, "Cantilever beam designs for RF MEMS switches," J. Micromech. Microeng. 20 (2010) 075042